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Standard Treatment Workflow (STW)

FEEDING AND NUTRITION IN NEONATES

ASSESS PHYSIOLOGICAL STABILITY FOR ENTERAL FEEDING

LOOK FOR: Shock / poor perfusion · Recurrent apnoea requiring intervention / respiratory failure · Ongoing seizures with physiological instability · Severe hypothermia · Concerning abdominal signs: bilious vomiting, bloody stools, marked or progressive distension, tenderness or discoloration

UNSTABLE / CONTRAINDICATION PRESENT

- Stabilize and treat the cause
- Start appropriate IV fluids
- Reassess clinical stability and readiness to start enteral feeds

STABLE / NO FEEDING CONTRAINDICATION

- Start / continue enteral feeds at the earliest
- Continue the tolerated feed volume in referred babies
- Choose route by gestational maturity and clinical status

TOTAL DAILY FLUID REQUIREMENT

ENTERAL + IV + OTHER FLUID INPUTS
Starting guide (mL/kg/day): Use day of life (D), not day of admission

GROUP	D1	D2	D3	D4	D5	D6	D7
BW <1000 g OR GA <28 wk	80	90	100	120	140	150	150–160
BW ≥1000 g AND GA ≥28 wk	60	80	100	120	140	150	150–160

Maintenance IV fluid type
First 48 h: 10% dextrose
After 48 h: Review serum Na, urine output and weight change before adding sodium. N/5 in 10% dextrose is an option; adjust for IV volume and enteral intake
KCl: usually 2 mmol/kg/day; review urine output, serum K and other K intake first

IV REVIEW → TRANSITION TO FEEDS

- Adjust total fluids and electrolytes as per daily weight change, urine output, blood glucose and clinical status
- **IV volume = total fluids − feeds − other fluid inputs**
- Start feeds after stabilization; taper IV as feeds progress; maintain hydration / glucose

MILK SOURCE

Preferred milk, in order:
Mother's own milk (MOM) → Pasteurized donor human milk → Appropriate formula if human milk unavailable

- Start expression within 1 h of birth; repeat every 2–3 h
- Use LMU for expression support, safe collection, storage and timely provision of MOM
- **Colostrum care:** Apply mother's own colostrum to buccal mucosa every 2–4 h when oral feeding is not possible

ROUTE OF FEEDING

<32 weeks	Orogastric (OG) tube feeds; encourage non-nutritive suckling at the expressed breast when stable
32 to <34 weeks	Spoon/paladai feeds; offer breast
≥34 weeks	Direct breastfeeding; supplement by spoon/paladai if intake is inadequate

Gestation/PMA is a guide. Use OG feeds if breastfeeding or spoon/paladai feeding causes coughing, choking, apnoea/desaturation, excessive fatigue or inadequate intake.

ENTERAL FEEDING PLAN: START AND ADVANCE

	STANDARD START	CAUTIOUS START: AEDF/REDF	CAUTIOUS START: VERY SMALL INFANTS
APPLIES TO	• GA ≥28 wk AND BW ≥1000 g, without antenatal AEDF/REDF • GA ≥35 wk, including antenatal AEDF/REDF†	GA 28–34⁶ wk AND BW ≥1000 g, with antenatal AEDF/REDF	GA <28 wk OR BW <1000 g
Start time*	As soon as possible after birth	As soon as stable; aim by 12 h	As soon as stable; aim by 12 h
Start volume	60 mL/kg/day	20 mL/kg/day	10–20 mL/kg/day
Advance daily as tolerated	20–30 mL/kg/day	20–30 mL/kg/day	20 mL/kg/day

If BW <1000 g, use the third column irrespective of gestation.
*Start once clinically stable, with reassuring abdominal assessment.
†At ≥35 wk, AEDF/REDF alone does not require restricted feeds or IV fluids; individualize by clinical and abdominal assessment.
FULL FEEDS: usually 150–180 mL/kg/day; individualize to growth and clinical status
FREQUENCY: 2-hourly; consider 3-hourly when PMA >32 wk, stable, on full feeds and off respiratory support
VOLUME PER FEED: daily enteral volume (mL) ÷12 (2-hourly) or ÷8 (3-hourly)

MONITORING

WHEN	WHAT TO ASSESS
Before each feed	Clinical condition, vomiting and visible abdominal distension; verify feeding-tube position when used
Daily	Weight change, actual milk intake and urine output; glucose / electrolytes as indicated
Weekly	Plot weight, length and head circumference on INTERGROWTH-21st postnatal growth charts
Immediately if warning signs	• Abdominal signs: bilious vomiting, bloody stools, marked/progressive distension, tenderness/discoloration • Any systemic clinical deterioration

Do not routinely check pre-feed gastric residuals or abdominal girth in well babies without signs of feeding intolerance.
Measure girth serially when there is distension or clinical concern.

POOR WEIGHT GAIN

Assess: actual milk intake/transfer, thermal stress, illness, anaemia and possible sodium deficiency
Optimize: correct identified causes; consider 180–200 mL/kg/day if tolerated and no fluid restriction
Review nutrient adequacy: Consider fortification if protein, energy or mineral intake remains insufficient

SUPPLEMENTS

SUPPLEMENT	WHO NEEDS IT	DOSE	START / REVIEW
Iron	Human-milk-fed preterm OR LBW infants	Elemental iron: BW <1500 g: 2–3 mg/kg/day BW 1500–2499 g: 2 mg/kg/day	Start at 2 weeks of age once feeds are established. Continue to 6 months, then follow Anemia Mukt Bharat.
Vitamin D	Consider in human-milk-fed preterm/LBW infants	400–800 IU/day As per unit protocol	Start once feeds are established; review at discharge and follow-up.
Calcium / phosphorus	Prioritize BW <1500 g OR GA <32 wk on unfortified human milk; others by intake / bone-disease risk	Preterm total intake targets: Ca 120–200; P 70–115 mg/kg/day Supplement the shortfall	Monitor P / ALP in at-risk infants. Review at term PMA; continue if intake is inadequate or tests remain abnormal.

Count nutrients from milk, fortifier and medicines; avoid duplication.
Consider Zinc in human-milk-fed preterm or LBW infants.

DOs

- Prioritize mother's own milk; help mothers express and breastfeed
- Start enteral feeds at the earliest; advance rapidly as per tolerance
- Review actual intake, growth and total fluids daily

DON'Ts

- Do not delay feeds without a clinical contraindication
- Do not routinely check gastric residuals or abdominal girth
- Do not overlook bilious vomiting, bloody stools or abdominal signs

KEY PERFORMANCE INDICATORS (KPIs)

Timely breastmilk feeding (%) — Target ≥80%

$$\frac{\text{Preterm/LBW neonates requiring facility-based care who received breastmilk within 12 h of birth (inborn) or admission (outborn)}}{\text{All preterm/LBW neonates requiring facility-based care}} \times 100$$

Exclusive breastmilk feeding at discharge (%) — Target ≥90%

$$\frac{\text{Preterm/LBW neonates admitted to SNCU/NICU who received exclusive breastmilk in the final 24 h before discharge}}{\text{Eligible preterm/LBW neonates discharged alive from SNCU/NICU}} \times 100$$

EARLY BREASTMILK FEEDING AND RAPID ADVANCEMENT IMPROVE SURVIVAL AND GROWTH

RELATED INFORMATION

Expressing the first milk

Cup / paladai feeding

Feeding low-birth-weight babies
NNF clinical practice guideline

ABBREVIATIONS

AEDF: Absent end-diastolic flow
ALP: Alkaline phosphatase
BW: Birth weight
D: Day of life
GA: Gestational age
IV: Intravenous
KCl: Potassium chloride
LBW: Low birth weight
LMU: Lactation management unit
MOM: Mother's own milk
NICU: Neonatal intensive care unit
OG: Orogastric
PMA: Postmenstrual age
REDF: Reversed end-diastolic flow
SNCU: Special newborn care unit

REFERENCES

1. WHO. Recommendations for care of the preterm or low-birth-weight infant. 2022.
2. Embleton ND et al. Enteral Nutrition in Preterm Infants (2022): ESPGHAN Position Paper. J Pediatr Gastroenterol Nutr. 2023;76:248–268.

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